



Audittens

ZKsync OS Security Review (v0.1.0...v0.3.0)

May 6, 2026

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1 About Audittens

Audittens is an audit company with expertise across various sectors of Web3 security.

What truly sets us apart is that our team consists entirely of top participants in mathematics and competitive programming competitions, uniquely equipping us to rapidly analyze codebases, uncover explicit and implicit invariants, and identify unique attack vectors.

You can subscribe and learn more about us at <https://x.com/Audittens>.

2 About ZKsync OS

ZKsync OS is a system-level implementation for ZKsync's state transition function.

Under ZKsync's new architecture, execution is decoupled from proving. ZKsync OS acts as the operation layer in this new architecture. It takes block data and an initial state as input and computes the new state after the application of the block. ZKsync OS is implemented as a Rust program.

You can learn more about the project at <https://zksync.io>.

3 Risk classification

The severities of the issues are determined based on the following properties:

- Critical — results in a significant loss of assets within the protocol, a major deviation from the expected behavior of protocol components, and/or a violation of key security invariants.
- High — causes loss of assets within the protocol and/or general violations of protocol security invariants, with limitations on the variety of potential attacks.
- Medium — enables barely profitable attacks on the protocol, violations of security invariants that pose minimal risk to protocol users, and/or functionality limitations affecting only a relatively small subset of users.
- Low — enables griefing attacks on the protocol, unexpected changes in the protocol's behavior that are imperceptible, and/or functionality limitations with negligible impact on the security of the protocol.
- Informational — issues that have no practical impact on the security of the current version of the protocol but could potentially lead to more serious consequences in future updates, create the possibility for incorrect use of the protocol by users, and/or result in a significant decline in code quality, making maintenance more challenging.

While determining the severity of issues, the constraints required for successful attacks and the potential actions to address their consequences are taken into account in the decision-making process.



4 Executive summary

4.1 Engagement overview

ZKsync Security Council Foundation engaged Audittens to review the security of components of ZKsync OS's implementation. Between December 1, 2025 and May 6, 2026, a team of 3 security researchers reviewed the provided source code over a total of 11 weeks within this period. After the completion of the fix period by ZKsync's team, the full security review was finalized by reviewing the corresponding commits.

4.2 Scope

Security review for the components of the `matter-labs/zksync-os` repository was provided.

All changes introduced by the [be15cc07...12833202](#) (v0.1.0...v0.3.0) diff, except for the changes related to Ethereum STF, were within the scope of the security review. Specifically, the following directories were excluded from the scope:

- `basic_bootloader/src/bootloader/block_flow/ethereum`,
- `basic_bootloader/src/bootloader/transaction_flow/ethereum`,
- `basic_system/src/system_implementation/ethereum_storage_model`.

4.3 Summary of findings

Severity \ Status	Status		
	Acknowledged	Fixed	Total
Critical	0	1	1
High	0	4	4
Medium	1	4	5
Low	4	13	17
Informational	7	38	45
Total	12	60	72

Table 1. Distribution of found issues.



5 Limitations

The security review was provided as is. Any subsequent changes may introduce new vulnerabilities and require a separate security review. Fixes for all findings have been reviewed within the limited scope (in the context of only relevant protocol components).

